

Cardinals

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Cardinals

Cardinality

Cardinal numbers capture the notion of the **cardinality** of a set. We measure the cardinality of a set relative to other sets by saying sets a, b have the **same cardinality** if there exists a bijection between them.

We use the following notation:

$$\begin{aligned} a \sim b & :\Leftrightarrow \text{there exists a bijection } a \leftrightarrow b \\ a \preceq b & :\Leftrightarrow \text{there exists an injection } a \hookrightarrow b \\ a \prec b & :\Leftrightarrow a \preceq b \wedge a \not\sim b \end{aligned}$$

It is straightforward to show $\sim$ is an equivalence relation and that $\preceq$ is transitive. $\preceq$ is obviously also reflexive. Relations that are reflexive and transitive are called **preorders**. If we mod out by $\sim$, do we get a partial order? In other words, do we have antisymmetry in the form

$$a \preceq b \wedge b \preceq a \stackrel{?}{\Rightarrow} a \sim b.$$

This follows pretty easily if we use the Axiom of Choice (in form of the well-ordering principle WO), but it is one of the few fundamental results in the theory of cardinality one can prove without using Choice.

Theorem 0.1 (Cantor-Schröder-Bernstein Theorem).

Let a and b be sets. If there is an injection from a to b and an injection from b to a , then there exists a bijection between a and b .

Exercise 0.1 (Warmup). Find a bijection between $[0, 1)$ and $[0, 1]$.

Proof. Let $X_0 = X \setminus g(Y)$, $Y_0 = Y \setminus f(X)$. Let $X_1 = g(Y) \setminus g(f(X))$, $Y_1 = f(X) \setminus f(g(Y))$. Note that there is a bijection between $X_0 \cup X_1$ and $Y_0 \cup Y_1$ by letting

$$\varphi(x) = \begin{cases} f(x) & \text{if } x \in X_0, \\ g^{-1}(x) & \text{if } x \in X_1. \end{cases}$$

Iterate this process with new $X := g(f(X))$ and $Y := f(g(Y))$. This gives rise to sequences of pairwise disjoint sets $X_0, X_1, X_2, \dots$ and $Y_0, Y_1, Y_2, \dots$, together with a bijection

$$\varphi : \bigcup_{i \in \omega} X_i \leftrightarrow \bigcup_{i \in \omega} Y_i.$$

We now define

$$h(x) = \begin{cases} \varphi(x) & \text{if } x \in \bigcup_{i \in \omega} X_i, \\ f(x) & \text{if } x \in X \setminus \bigcup_{i \in \omega} X_i. \end{cases}$$

It is clear that h is one-to-one. Suppose $y \in Y \setminus \bigcup_{i \in \omega} Y_i$ is not in $f(X \setminus \bigcup_{i \in \omega} X_i)$. But then, by definition of φ , y is not in $f(Y)$, which would imply $y \in Y_0$, contradiction. Hence h is onto. $\square$

The next result shows that this order is linear. To prove it, we have to use Choice.

Theorem 0.2 (Hartogs).

For any sets a, b ,

$$a \preceq b \vee b \preceq a.$$

Proof. By the well-ordering principle WO, there exist ordinals α, β such that $a \sim \alpha$ and $b \sim \beta$. Since any two ordinals are comparable by $<$, $\alpha \subseteq \beta$ or $\beta \subseteq \alpha$, which easily yields the theorem. $\square$

Cantor's famous result shows that for every set there exists one with greater cardinality. In particular, the order $\preceq$ is infinite. Given a set a , we denote its **power set** by $\mathcal{P}(a)$,

$$\mathcal{P}(a) = \{b : b \subseteq a\}.$$

Theorem 0.3 (Cantor's Theorem).

For every set a ,

$$a \prec \mathcal{P}(a).$$

Proof. $x \mapsto \{x\}$ is an injection from a to $\mathcal{P}(a)$, hence $a \preceq \mathcal{P}(a)$. Now assume $a \sim \mathcal{P}(a)$ via a bijection f . We use a diagonal argument (as for the case $a = \mathbb{N}$) to derive a contradiction. Let $d := \{x \in a : x \notin f(x)\}$. As f is onto, there exists $b \in a$ with $f(b) = d$. But then

$$b \in d \Leftrightarrow b \notin f(b) \Leftrightarrow b \notin d,$$

contradiction. $\square$

Cardinal numbers

As with ordinals, we would like to choose a system of representatives for the equivalence classes under $\sim$. Theorem 0.2 suggests to use ordinals (which we can compare using the well-ordering of ordinals). We therefore assume the Axiom of Choice (so every set has a well-ordering). One issue that remains is that a set can have well-orderings of different lengths (as we saw in the case of the natural numbers $\mathbb{N}$). But since the ordinals are well-ordered, we can choose the *least* one of the same cardinality.

Definition 0.1.

The **cardinality** $|a|$ of a set a is defined as

$$|a| = \min\{\alpha \in \text{Ord} : \alpha \sim a\}.$$

Remark 0.1. We have to be careful what we are taking a minimum over, since there could be a lot (i.e. “more than a set” many ordinals equinumerous to a .) We can avoid this by first using AC to find *one* ordinal $\beta \sim a$, and then taking the minimum ordinals $\leq \beta$. As we will see later, the latter will always be a set.

Lemma 0.1.

- $a \sim b \iff |a| = |b|$
- $a \preceq b \iff |a| \leq |b|$
- $a \prec b \iff |a| < |b|$

An ordinal κ is a **cardinal** if it is the cardinality of some set. Equivalently, κ is a cardinal if

$$\forall \beta < \kappa \ \beta \prec \kappa.$$

We usually use Greek letters from the middle of the alphabet, $\kappa, \mu, \nu, \dots$ to denote cardinals.

Given a cardinal κ , we let κ^+ be the **cardinal successor** of κ , i.e.

$$\kappa^+ = \text{least cardinal } \nu \text{ with } \kappa \prec \nu.$$

By Theorem 0.3 (and the Axiom of Choice), we have

Remark 0.2. The fact that $\kappa^+ \preceq |\mathcal{P}(\kappa)|$ also ensures in the definition of κ^+ we are taking the minimum over a set.

$$\kappa \prec \kappa^+ \preceq |\mathcal{P}(\kappa)|,$$

and as ordinals,

$$\kappa < \kappa^+ \leq |\mathcal{P}(\kappa)|.$$

All finite ordinals

$$0, 1, 2, 3, \dots$$

are cardinals. The **first infinite cardinal** is ω , the cardinality of all countably infinite sets.

We can use ordinals to enumerate all infinite cardinals. This gives the **aleph**-sequence.

$$\begin{aligned}\aleph_0 &:= \omega \\ \aleph_{\alpha+1} &:= \aleph_\alpha^+ \\ \aleph_\lambda &:= \sup\{\aleph_\beta : \beta < \lambda\}\end{aligned}$$

Exercise 0.2. Show that the supremum of a set of cardinals is a cardinal. (In particular, $\aleph_\lambda$ in the definition above is a cardinal.)

As in the case of ordinals, cardinals of the form $\aleph_{\alpha+1}$ are called **successor cardinals**, whereas alephs whose index is a limit ordinal are called **limit cardinals**.

Proposition 0.1.

Every cardinal is an aleph, i.e. for every cardinal κ there exists an ordinal α such that $\kappa = \aleph_\alpha$.

Sketch. The mapping $\alpha \mapsto \aleph_\alpha$ is strictly increasing. It follows that $\kappa \leq \aleph_\kappa$ and hence $\kappa < \aleph_{\kappa+1}$. Choose α minimal such that $\kappa < \aleph_\alpha$. A simple case distinction yields that α must be a successor ordinal, $\alpha = \beta + 1$. It then follows from $\aleph_{\beta+1} = \aleph_\beta^+$ that $\kappa = \aleph_\beta$. $\square$

While every cardinal is an ordinal, not every ordinal is a cardinal. For example

$$\aleph_0 = \omega = |\omega + 1| = |\omega + 2| = \dots = |\omega + \omega| = |\mathbb{Z}| = |\mathbb{N} \times \mathbb{N}| = |\mathbb{Q}|.$$

The next cardinal is

$$\aleph_1 = \aleph_0^+ = \omega_1 = \{\alpha \in \text{Ord} : \alpha \text{ countable}\}.$$

Between $\aleph_0$ and $\aleph_1$ are uncountably many ordinals, all of which are countable. The ordinal ω_1 will play an important role when we close sets under countably infinite operations (like the Borel sets).

We know the set of reals is uncountable, so at this time we are able to conclude

$$|\mathbb{R}| \geq \aleph_1, \quad |\mathcal{P}(\mathbb{R})| \geq \aleph_2, \dots$$

Operations on cardinals

What can we say about the cardinality of unions, products, etc.?

Exercise 0.3. Show that $|\mathbb{R} \times \mathbb{R}| = |\mathbb{R}|$.

This is an instance of a more general result. We can define arithmetic operation on cardinals by looking at the cardinalities of the corresponding set theoretic operations.

$$\begin{aligned} \kappa + \lambda &= |(\kappa \times \{0\}) \cup (\lambda \times \{1\})| && \text{(disjoint union)} \\ \kappa \cdot \lambda &= |\kappa \times \lambda| && \text{(product)} \\ \kappa^\lambda &= |\{f : f : \lambda \rightarrow \kappa\}| && \text{(exponentiation)} \end{aligned}$$

Exercise 0.4. Verify that for finite cardinalities, the operations defined above agree with the usual arithmetic operations on natural numbers.

It turns out for infinite cardinals, $+$ and $\cdot$ are trivial. This is mostly due to the fact that there is a nice (canonical) well-ordering of $\text{Ord} \times \text{Ord}$.

The canonical well-ordering of ordinal pairs

We define a bijection $F : \text{Ord} \times \text{Ord} \leftrightarrow \text{Ord}$. As a first step, we define the **lexicographic order** $<_{\text{lex}}$ on $\text{Ord} \times \text{Ord}$:

$$(\alpha, \beta) <_{\text{lex}} (\gamma, \delta) \quad : \iff \quad \alpha < \gamma \vee (\alpha = \gamma \wedge \beta < \delta).$$

This is a linear order in which every non-empty subset has a minimal element, but we have to be careful since in this order all pairs of the form $(0, \xi)$, $\xi \in \text{Ord}$, precede $(1, 0)$. And we saw that there is no set of the form $\{(0, \xi) : \xi \in \text{Ord}\}$. Gödel modified this order by *presorting* according to the maximum of the pair:

$$(\alpha, \beta) <_g (\gamma, \delta) \quad : \iff \quad \max(\alpha, \beta) < \max(\gamma, \delta) \tag{1}$$

$$\vee (\max(\alpha, \beta) = \max(\gamma, \delta) \wedge (\alpha, \beta) <_{\text{lex}} (\gamma, \delta)). \tag{2}$$

This yields a linear order in which every element has only a *set* of predecessors. Moreover, the minimality condition is still satisfied: Let $A \subseteq \text{Ord} \times \text{Ord}$ be non-empty. We find the least element of A by - first finding the least $\gamma_0 \in \{\max(\alpha, \beta) : (\alpha, \beta) \in A\}$, - then finding the least $\alpha_0 \in \{\alpha : \exists \beta ((\alpha, \beta) \in A \wedge \max(\alpha, \beta) = \gamma_0)\}$, - and finally finding the least $\beta_0 \in \{\beta : (\alpha_0, \beta) \in A \wedge \max(\alpha_0, \beta) = \gamma_0\}$.

It is straightforward to verify that (α_0, β_0) is the smallest element of A with respect to $<_g$.

Enumerating pairs of ordinals along $<_g$ now yields an order isomorphism $F : \text{Ord} \times \text{Ord} \rightarrow \text{Ord}$,

$$(\alpha, \beta) <_g (\gamma, \delta) \iff F(\alpha, \beta) < F(\gamma, \delta).$$

Note that $f(\alpha) := F(\alpha \times \alpha)$ is a strictly increasing map on ordinals, and hence $F(\alpha) \geq \alpha$. Further note that

$$\alpha \times \alpha = \{(\beta, \gamma) : (\beta, \gamma) <_g (0, \alpha)\},$$

hence $\alpha \times \alpha$ is a $<_g$ -initial segment.

An important application of the well-ordering of $\text{Ord} \times \text{Ord}$ is the following theorem.

Theorem 0.4 (Hessenberg's Theorem).

Let κ be an infinite cardinal. Then

$$\kappa \cdot \kappa = \kappa.$$

Proof. Show that for any infinite cardinal κ , the restriction of the mapping F to $\kappa \times \kappa$, $F|_{\kappa \times \kappa}$, is a bijection between $\kappa \times \kappa$ and κ . In other words, show that $F(\kappa \times \kappa) = \kappa$.

It is straightforward to verify this for $\kappa = \aleph_0$.

Now assume it does not hold for a larger cardinal. Since every infinite cardinal is of the form $\kappa = \aleph_\alpha$, choose $\alpha > 0$ minimal such that $F(\aleph_\alpha \times \aleph_\alpha) \neq \aleph_\alpha$.

Note that since for any β , $F(\beta, \beta) \geq \beta$, we must have $F(\aleph_\alpha, \aleph_\alpha) > \aleph_\alpha$.

Since F is a bijection, there exist ordinals $\beta, \gamma < \aleph_\alpha$ such that $F(\beta, \gamma) = \aleph_\alpha$. If we put $\delta = \max\{\beta, \gamma\} + 1$, we have $\omega \leq \delta < \aleph_\alpha$ (since $\aleph_\alpha$ has to be a limit ordinal).

Since $(\beta, \gamma) <_g (\delta, \delta)$, it follows that $F(\delta \times \delta) > \aleph_\alpha$ and thus $F(\delta \times \delta) \supseteq \aleph_\alpha$. The latter implies $|\delta \times \delta| \geq \aleph_\alpha$. But $|\delta \times \delta| = |\delta| \cdot |\delta|$ and by minimality of α , $|\delta| \cdot |\delta| = |\delta| < \aleph_\alpha$, contradiction. $\square$

Corollary 0.1.

If κ, λ are infinite cardinals, then

$$\kappa + \lambda = \kappa \cdot \lambda = \max\{\kappa, \lambda\}.$$

Proof. If κ is infinite, we have, by simple monotonicity arguments,

$$\kappa \leq \kappa + \kappa \leq \kappa \cdot \kappa = \kappa, \quad \text{hence } \kappa + \kappa = \kappa \cdot \kappa = \kappa.$$

If, say, $0 < \kappa \leq \lambda$ with λ infinite, the above implies

$$\lambda \leq \kappa + \lambda \leq \lambda + \lambda = \lambda, \quad \text{and thus } \kappa + \lambda = \lambda = \max\{\kappa, \lambda\}, \quad (3)$$

$$\lambda \leq \kappa \cdot \lambda \leq \lambda \cdot \lambda = \lambda, \quad \text{and thus } \kappa \cdot \lambda = \lambda = \max\{\kappa, \lambda\}. \quad (4)$$

$\square$

The cardinality of $\mathbb{R}$ and related sets

While, as we just saw, addition and multiplication are trivial for infinite cardinals, determining the simplest transfinite power, $2^{\aleph_0}$, already leads to unsolvable problems. $2^{\aleph_0}$ plays an important role since it is the cardinality of the continuum.

Proposition 0.2.

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0}.$$

Hint

Use, for example:

- Dedekind cuts
- every real number has an infinite binary expansion
- subsets can be identified with their characteristic function

The Theorem 0.1 makes life easier here.

We can draw some conclusions:

1. With $|\mathbb{R}| = 2^{\aleph_0}$ we also get $|\mathbb{R}^n| = (2^{\aleph_0})^n = 2^{\aleph_0}$, by Hessenberg's Theorem. It even holds that (exercise!)

$$|\{s: \mathbb{N} \rightarrow \mathbb{R}\}| = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0}.$$

That is, there are as many countable sequences of real numbers as there are real numbers.

2. Since every continuous real-valued function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$, we have

$$|\{f: \mathbb{R} \rightarrow \mathbb{R} \text{ continuous}\}| = |\{f: \mathbb{Q} \rightarrow \mathbb{R}\}| = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}.$$

3. We obtain a higher cardinality by passing to the power set of the real numbers or to the set of all real-valued functions on $\mathbb{R}$:

$$|\{f: \mathbb{R} \rightarrow \mathbb{R}\}| = |\mathcal{P}(\mathbb{R})| = 2^{2^{\aleph_0}} > 2^{\aleph_0}.$$

Continuum Hypothesis (CH – cardinal version)

$$2^{\aleph_0} = \aleph_1$$

CH asserts that the cardinality of $\mathbb{R}$ is the next biggest cardinal after $\omega = \aleph_0$ (the cardinality of all countably infinite sets).